

CS310 Project Progress Report

Mobile Application Enhancing Transparency of UK Politics

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1 Introduction

The emergence and subsequent growth of new Natural Language Processing (NLP) technologies has allowed for new developments in analysing and interpreting texts. This has a broad scope of applications, one such being is enhancing the transparency between governments and their people. As discussed in the project specification, legislative texts are difficult to read and since the average reading age of the UK population has been estimated to be 9 years[48, 38], it demonstrates that legislation is far from easy for the average reader to understand. Using NLP technologies of Large Language Models (LLMs) and transformers, we can create a solution.

1.1 Problem

The average population of countries lack the ability to easily read, understand and stay up to date with developments in their country's governance. This particularly applies to laws, acts, bills and other legislative texts that are not easy to interpret, demonstrated across several papers[11, 5, 7]. This project aims to automate simplifying UK legal texts to be more readable.

FKGL	ARI	FK	GFI	SMOG	Word count	Words\sentence
16.26	19.02	41.83	21.22	16.56	11972.50	35.57

Figure 1: Average of relevant metrics of 631 UK legal documents from between 1998 and 2023, which contain over 500 words.

Parsing a sample of UK legal documents and calculating their [readability metrics](#)[8], we can see that the average number of years of education required to read these texts is 16, far above the average reading age. Furthermore with a large average word count (exceeding that of a novel[41]) and overly long sentences[21] that UK legal texts are far from easy to read. These laws impact so many people's lives - within and outside of the UK's borders - and it is important that the people have some way of understanding what changes their government are making. While some systems do exist to increase transparency[50], we can use developments in NLP to solve this problem.

1.2 Objectives

The ultimate aim of this project is to create a product that facilitates transparency between the population of the UK, and the government. This would be achieved by creating a mobile application, which would perform two crucial tasks: keeping relevant people and parties up to date on the bills and laws which may impact them; and simplifying those bills and laws to their principal components, so they can be more easily understood. These objectives are fairly difficult to achieve within our time frame; hence this aims to build the framework which the project can then expand from. Within my specification, we outlined 7 core modules which defined the objectives.

1. **Text simplification:** simplifying the given bill or act.
2. **Political sentiment analysis:** determining if the given output is biased.
3. **Live updating system:** updating and notifying relevant users of recently proposed bills.
4. **Demographic identification:** identifying the most relevant demographics to a proposed bill, and how they are impacted.
5. **Database:** space efficient and secure database which stores relevant details of users.
6. **UI:** intuitive and easy to use user interface to navigate bills and acts.
7. **Ethical discussion:** exploring the potential risks, drawbacks and implications of producing this application.

Within this document, we outline the progress that has been made in the two months since beginning this project. It will describe the research undertaken, the state of the project, how close we are to achieving the set objectives, and how the project will need to adapt to developments made.

2 Research

The initial phase of this project has been targeted toward research. The priority has been learning, understanding and evaluating methods we can use to solve our problem - particularly focusing on interpreting structures of legal texts, ML, NLP techniques and any existing solutions.

2.1 UK legislation and bills

It was important to learn to understand the basics of UK law, legislation and bills. Understanding the fundamentals of law creation and structure was the first step. All information gathered was through the UK government website[36].

2.1.1 Introducing bills

A bill is a proposal for a new law, or a proposal to change an existing law. Different types of bills can be introduced by the government, individual MPs or Lords, and private individuals or organisations, resulting in differently structured documents. They are introduced in House of Commons or House of Lords for examination, discussion and amendment. When both houses have agreed on the content of a bill, it is presented to the reigning monarch for ascension, then it becomes a law. This process is important because at each stage it specifies what we know about the bill, and our application will need to be able to pull all relevant information to give an adequate summary of it. In the earlier stages such as the first reading, much of the information is not found in the published document but found in the parliamentary debate, which we can access through Hansard[34].

2.1.2 Legislation structure

The structure of a legal text is irregular, it does not follow general sentence or paragraph structures. These texts include long lists; sometimes connected with “and”s, other times written as to complete a previous sentence. Bills are an early form of an act, which evolves and develops into the final legal document. The common structure of the bill is reflective of the structure of an act, which is important to reading the texts. Bills and acts can have varying levels of detail, but generally keep to this structure:

- **Title:** Identifies its purpose and scope. It is a concise description of the bill.
- **Preamble:** Provides a brief introduction and explanation of the reasons for introducing the bill. Outlines the goals and objectives the bill aims to achieve.
- **Long Title:** More detailed version of the title, specifies the main purpose and intent of the bill.
- **Short Title:** Abbreviated name for the bill.
- **Cause-By-Cause Provisions:** Main body of the bill consists of a series of clauses each addressing a specific aspect of the proposed law. They are numbered sequentially. These clauses detail the changes or additions to existing legislation or lay out the new legal framework being proposed.
- **Schedules:** Attached to the bill and provide additional details such as lists, tables, or complex provisions that support the main clauses. Schedules are usually referred to by name, or number within clauses of the bill.
- **Enacting Formula:** Formal statement that indicates the authority under which the bill is being passed. It typically takes the form, “Be it enacted by the Queen’s Most Excellent Majesty...”
- **Commencement:** Outlines when the different provisions of the bill may come into effect. Some may be immediately upon royal assent, others might have a delayed commencement date.
- **Extent:** Defines geographical scope of the legislation, defining where in the UK the bill applies to.
- **Interpretation:** Section defines key terms used within the bill to avoid ambiguity or confusion.
- **Annotations:** Explanatory notes that provide additional context or clarification.

2.2 Readability metrics

Readability metrics are numerical measures of features of a text. This often refers to complexity of a text, and how difficult it is to read. However there are other relevant components such as how well meaning is carried between two texts and lexical complexity. There are a large host of metrics to consider that may be useful, however to our project specific task, its important to narrow down which metrics are relevant.

2.2.1 Reading ease

Flesch Kincaid Grade Level[17] (FKGL):

A widely used readability metric, which tells us an approximation of the USA grade level, the required reading grade to understand the text. This can be understood also as the number of years of education needed to read the text¹. This metric is largely used in education, so that the text of the reading age can be deemed appropriate.

$$\text{FKGL} = 0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59$$

Flesch Kincaid Reading Ease[17] (FK):

This is the most widely used metric in looking at reading ease. It gives a numerical score where the smaller the value, the more difficult it is to read. This metric is used in many places, such as in word processing, US Department of Defense and other industries.

$$\text{FK} = 206.835 - 1.015 \left(\frac{\text{total words}}{\text{total sentences}} \right) - 84.6 \left(\frac{\text{total syllables}}{\text{total words}} \right)$$

Gunning Fog Index[42] (GFI):

This metric is used to ensure the text is readable by its target audience. It gives an approximation of the number of years of education the readers need to have. It is described as being good at determining difficult texts, and generally requires below 12 to be applicable to a general audience.

$$\text{GFI} = 0.4 \left(\left(\frac{\text{total words}}{\text{total sentences}} \right) + 100 \left(\frac{\text{complex words}}{\text{words}} \right) \right)$$

Simple Measure of Gobbledygook[25] (SMOG):

Another widely used metric, as an improvement to GFI. It similarly gives an approximation of the number of years of education the readers need to have. It is used to improve health-care messages, so they are readable by all patients.

$$\text{SMOG} = 1.0430 \sqrt{\text{total polysyllables} \times \frac{30}{\text{total sentences}}} + 3.1291$$

These four metrics have been selected due to their wide acclaim, usage and variety between them in how they determine a complex word. We have chosen to disregard Dale Chall readability[12], the Raygor estimate[16] and Spache readability[43] for their limited scope beyond lower education. Automated Readability Index[45] (ARI) and Coleman Liau Index[23] (CLI) do see some use in research, but were designed with older technology in mind, making other options more favorable.

2.2.2 Text similarity

We want to increase the reading ease of our texts, but if we were to just use for example Flesch reading ease, our text would be simplified in ways that may lose meaning of the original text[47]. Hence we need to ensure meaning is maintained between texts, through precision and recall.

Recall-Oriented Understudy for Gisting Evaluation: [14] (ROUGE)

This is most commonly used as a metric for evaluating text generation models, measuring the overlap between text and the given model prediction. There are several variants each emphasising a certain attribute. ROUGE-N is the most commonly used, and is the direct n-gram (contiguous sequence of n items from a given sample of text) overlaps between the text and the prediction. Scores lie within [0, 1] such that 1 indicates a total match between the text and prediction. ROUGE has a significant limitation of summaries using identical n-grams achieving high scores, making ROUGE worse at [abstractive summarisation](#)[28].

¹We can calculate this with respect to UK grade levels without much loss of generality.

$$\begin{aligned}\text{ROUGE-N}_{\text{recall}} &= \frac{|\text{text n-grams} \cap \text{prediction n-grams}|}{|\text{text n-grams}|} \\ \text{ROUGE-N}_{\text{precision}} &= \frac{|\text{text n-grams} \cap \text{prediction n-grams}|}{|\text{prediction n-grams}|} \\ \text{ROUGE-N}_{\text{Fmeasure}} &= 2 \cdot \frac{\text{ROUGE-N}_{\text{recall}} \cdot \text{ROUGE-N}_{\text{precision}}}{\text{recall} + \text{precision}}\end{aligned}$$

System output Against References and against the Input sentence:[53] (SARI) This metric compares how similar two texts are by considering a series of operations on a text: addition, keep and deletion. SARI calculate the sum of the weighted F-measure of each operation. The result lies in the range $[0, 100]$ range, with higher scores indicating better performance: greater simplicity and meaning retention. This metric was built from extending FK and Bilingual Evaluation Understudy (BLEU), and has great potential in being used for language models.

BERTScore: [54] This is an evaluation metric for NLP, which calculates the similarity between two texts by performing pairwise cosine similarity. It differs to the other methods by calculating similarity based on contextual embeddings, which is gathered by Bidirectional Encoder Representations from Transformers (BERT), a pretrained language model. These contextual embeddings carry weights which can give certain words higher importance upon evaluation. This may be something to be wary of, due to BERT’s sensitivity to its tokenization and lexicon.

These metrics have been looked at in more detail than alternatives such as BLEU and Metric for Evaluation of Translation with Explicit Ordering (METEOR), due to them being built for translation models over summarisation models.

2.3 Text simplification

To perform text simplification, we need to use solutions to text summarisation. Simplification and summarisation are two separate NLP tasks[14], with simplification having the goal to reduce the text to something anybody can read, whereas summarisation is to provide an overview of the text. We can approach summarisation either to extract or to abstract information[10, 24]. Extractive summarisation aims to isolate the most important and relevant quotes from a text, and pulling them out into a summary. Abstractive summarisation aims to rewrite the key topics into a coherent self contained passage, trying to best reflect the meaning of the text while introducing new text.

2.3.1 Lexical simplification

One aspect of complex texts is the abundance of complex words. This technique aims to replace complex words with simpler corresponding words. Often this is achieved given a dataset of word synonyms such as WordNet[1]; subsequently the most complex words are identified, selected and replaced. The LSBERT model[40, 39] is a powerful tool for many common texts to accurately identify complex words using BERT, linearly replacing them with simpler synonyms provided by the model. Legal documents may need additional data provided, using the UK’s Parliamentary glossary[33]. LSBERT is great at simplifying words, but may struggle with phrases unless we tokenize differently.

2.3.2 Syntactic simplification

Another aspect of complex texts are long sentences with large amounts of information contained in it. This technique is focused on reducing the grammatical complexity of a text, while retaining its meaning[44]. One method commonly explored is sentence simplification, as the context of a sentence is easily self contained. Due to our texts being long(1), sentence simplification alone is not enough. To perform simplification across the text, we can perform extractive summarisation[24] finding the cosine similarity of sentences and using the highest frequencies to identify the most important sentences and topics. We can then use abstractive summarisation on the extracted points to produce an easier to read summary than the input.

2.4 Machine learning techniques

To implement the discussed methods of text simplification, we will want to use an existing language model, so not to have to build one from the ground up. There are two core ML approaches for simplification we can take.

2.4.1 Supervised learning & fine-tuning

We can train a language model by giving it input text, and text we want it to transform into; and it will identify patterns it can then use on newer entries. Hugging Face uses transformers - a neural network which transforms an input - to perform and solve several NLP tasks. It comes prepared with tokenizers, datasets, existing models and an ability to fine tune models with new datasets. We can use the summarisation tutorial[9] as a guide to fine-tune an existing Sequence to Sequence (Seq2Seq) model. Hugging Face provides GPT-2, PEGASUS, T5 and BART as Seq2Seq models we can fine-tune.

Taking this approach we need pairs of legal text and their summaries. We can access large amounts of data from the UK Parliament APIs, but we lack the simplified text. Options to consider are using long titles in the legal document over a certain length (similar to the tutorial); using legal experts to summarise given passages[15]; using existing news articles which cover new legal developments; or using existing summarisation models to simplify the legal text, and we take the best summaries. Due to the provided Seq2Seq models having a maximum input of 1024 tokens, which far exceeds the average word count², we will need some method to parse the long texts. Due to being trained on ROUGE, extractive summarisation is where it will most succeed at; however Hugging Face also has other useful modules such as text classification and sentiment analysis that we can use.

2.4.2 Large language models

New LLMs which have been trained on massive swathes of data, have demonstrated emergent abilities[52]. This has shown to have many applications in NLP tasks such as summarisation, showcased by how it is being advertised for summarising articles and research papers for students[2, 26, 22]. Moreover, its summarisation capabilities are being used in finance[13] and medicine[20]. These are useful comparisons as they cover similarly complex fields to the legal domain. Additionally, this method has been used on a sample of UK bills with Padolsey's Parse The Bill project[31] which attempts to perform a similar task to what we are doing. LLMs are autoregressive models that using their previous inputs, to predict the next output that is to come, providing them with contextual knowledge in answering prompts, which can be used to extend their abilities. As the scale of an LLM grows, the LLM will perform better on certain NLP tasks[29]. We can understand why summarisation is so successful using an LLM because of the contextual information provided about specific subject matters, as seen with FinBERT[13], where LLMs trained with finance specific knowledge perform better than those without. The number of LLMs we can use has grown, with models such as Llama2, GPT-3.5, BERT, Bloom, Falcon180B etcetera. Notably Llama2 has recorded success on better summarising US bills better than GPT-4[30].

LLMs have already shown some success at simplifying legal documents[31] and have the advantage we do not need to train them. Additionally they are great at abstracting away irrelevant information like meaningless references and citations - which are not important for a summary. This data abstraction does bring a loss of accuracy[27, 31] which is difficult to control and monitor, resulting in certain [ethical concerns](#). LLMs share the token limit problem with transformer models (despite Llama2 and GPT3.5 having maximums of 4096 tokens) meaning we still need an approach to parsing long texts.

2.5 Parsing long texts

To use any ML technique we need to find some method to reduce the input size. We can truncate aspects that may not be relevant to the summary, such as the Schedule aspect of a bill. This alone would not be enough to enter the token range. A widely used approach is recursive summarisation. This is the process of chunking the text into logical sections (complete sentences), usually as large as possible. Then we extractively summarise (to minimize information loss) these chunks, before concatenating the summaries back together. If the summary is not concise enough, we can recursively repeat the process until our summarisation is our intended length or complexity. There is a large disadvantage that in sequential texts the context of the prior chunk will be lost, meaning certain topics will be undervalued in relevance. This can be accounted for in several methods:

- Isolate the first chunk, summarise the chunk, then concatenate it back to the start of the text. Repeat until entire text has been summarised.
- Take several overlapping chunks and use the most recurring topics to select which summaries to use.
- Extract every topic from each chunk separately, and weigh them against each other at the end to see which topics to prioritise in each summary [31].
- Use topic grouping on the text, then summarise each topic directly[49, 6].

We may also consider optimising this process, such as performing these operations on already tokenized data, and only translating back upon completion.

²Each token is not necessarily each word, but it is a viable approximation.

2.6 Conclusions

Using this research we can use these techniques to simplify our texts, combining them to optimize the output. Existing solutions have often used these methods in isolation, leading to certain flaws. Padolsey’s approach[31] of using LLMs carries the noticeable loss in accuracy. Meanwhile, papers utilizing transformers[5, 11, 15] were held back by the model not having been trained on a legal corpora. This resulted in their models lacking contextual information and an adequate legal lexicon. Using a combination of both transformers and LLMs, we can mitigate the flaws in each approach, to create an application which can better simplify legal texts.

3 Project overview

3.1 UK Parliament API

To access any data for simplification, we needed a method to acquire bills and acts in string form. Fortunately UK Parliament provides an API for accessing UK bills[32] and UK legislation[35]. Using both of these we can make a large dataset we can use for training, testing and validating.

To access passed UK acts, we can curl the link:

```
"https://www.legislation.gov.uk/ukpga/{Year}/{ID}/enacted?view=plain"
```

where the year it was published, and an ID act as a composite key. This returns to us a HTML file, where we look at the content that is a child of the “viewLegContents” element. We can parse this using the BeautifulSoup library in Python, and to get the raw text of everything, we can use the html2text library[46]. We can access legislation that coheres to this structure, all the way back to 1900. However the number of documents in the HTML format is much lower the further back we go. This gives us over 1000 legal documents that we can use already.

Gathering UK bills meanwhile is more difficult. The Bills API lets us find out all information about any bill, returning a JSON of relevant information, such as its sponsor, stage, last updated, etc. To access the actual documents concerning the bill, we first need to curl:

```
"https://bills-api.parliament.uk/api/v1/Bills/{Bill ID}/Publications"
```

where we pass in the **Bill ID**. This link will provide a JSON with a list of every publication associated with this bill (it can be specified to be ascending or descending by date). Each publication has a list of documents, and a list of links. The documents can be a PDF or a HTML, and will have an associated document ID. We can curl:

```
"https://bills-api.parliament.uk/api/v1/Publications/{Publication ID}/Documents/{Document ID}/Download"
```

which will give us the byte data of the downloaded file, which if it is a HTML, we can translate the same way we would legislation. Meanwhile, the links go to a Parliamentary website with the related bill content, which because it is not built with the API, automated access often results in the request being blocked by Cloudflare. We can get around this by Selenium, which may take more time. Each bill does not need to have a published document until the committee stage, giving us only 357 bills to work with. Gathering details from newly proposed bills may require more work.

When we extract the text as a pure string, the text given is flattened, meaning that there is no maintenance of indentation, and the text is disturbed. This format is usable[31], but comes with less control and brings several issues in how models may read and interpret the text. This is where we want to use an algorithm to unflatten the text, by converting into a tree, where the children of a node is the text indented beneath that node. We can gather the indentation level from the raw HTML document, or heuristically from the string. We can then convert the tree into a JSON reflecting the original document.

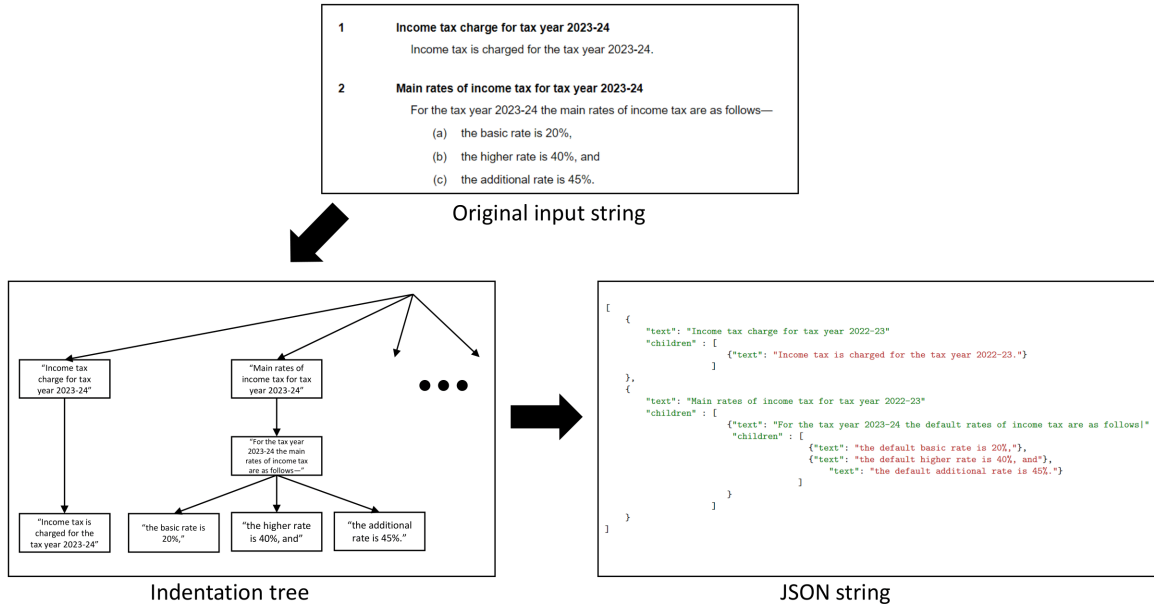


Figure 2: Process to convert text to discussed JSON format on an extract of the Finance No. 2 Act 2023.

3.2 Progress

Compared to the original timeline, the initial plan has not succeeded. The aim was to have module 1 completed, and for the database and UI to have started development. However, research took substantially more time than anticipated. Research has been more thorough and detailed, covering a wider span of topics and techniques, yet we have not produced a Minimum Viable Product (MVP) using our findings. This was a result of underestimating how much content I had to learn for ML; and the time taken accessing and parsing our datasets. While I am behind on every module, this setback is not as damaging to the project due to the redundancy provided by MoSCoW for each requirement. Certain extension features can be deprioritised in producing an MVP for the coming sprints. Furthermore, while research was completed under the pretences of module 1, the methods explored directly assist in several other modules - especially sentiment analysis and demographic identification. Additionally in writing code to generate our datasets, we have begun the design of the database to be used for the MVP. The research completed has laid the foundations to build the remainder of the project, allowing us to enter the development phase.

4 Project Evolution

4.1 Changes to plan

The original plan intended to produce a system with features from all seven modules. With the limited progress we have made, we now need to reconsider and deprioritise modules so that we can create a more successful project. The plan has now changed to change focus to creating a working MVP as soon as possible. Once this has been created, an evaluation period can be taken, where we can determine the success of our simplification tool and then expand it with additional features and modules if feasible. As seen on the new updated Gantt chart(3), the later weeks are left open for changes needed to existing modules, adapting from the success of each sprint.

Module	Changes
Module 1: Text simplification	Development will be greatly prioritised to be completed as soon as possible in the coming sprints. The simplification will aim for a 20% improvement in discussed readability metrics , compared to Padolsey’s solution[31].
Module 2: Political sentiment analysis	This will be deprioritised for implementation, rather we will use Hugging Face sentiment analysis libraries[37] to evaluate if for future development the app will need to account for significant or harmful bias.

Module 3: Live updating system	Deprioritised for MVP, as instead of setting up a server system to check repeatedly for updates, we will pretend we are looking at the system at a snapshot in time, and emulate a bill being updated in the system. This way it acts as a proof of concept, and live updates can be treated as a black box external to the system.
Module 4: Demographic identification	Deprioritised for MVP, however our techniques for summarisation blend into who is impacted, as seen with Padolsey[31]. Design will be developed alongside module 1, and implemented if feasible.
Module 5: Database	Highly prioritised for MVP, as this needs to be created in a minimal form. We will work with it as if there is a single user, and extend for more users in later sprints. To avoid GDPR issues, fictional users will be implemented.
Module 6: UI	Highly prioritised in a minimal form. This will be built alongside the database to ensure that when the MVP is deployed we can demonstrate the uses of the app, accommodating existing features. We will also set up dummy icons and pages where other features may eventually be implemented.
Module 7: Ethical discussion	Will continued to be explored, especially with respects to the developments in module 1 and 2. The results found on the success of the text simplification will be used to inform if the project is ethically viable, with a justification of the conclusion.

4.2 Changes to timeline

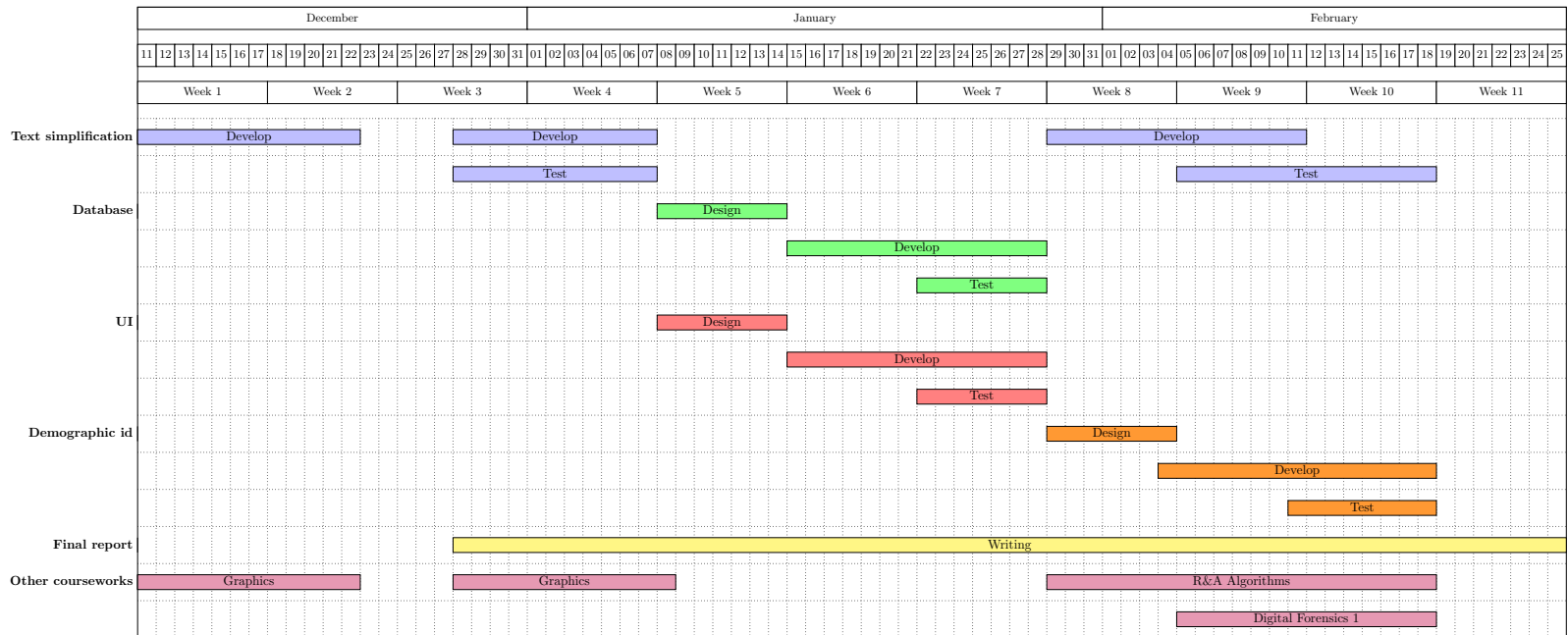


Figure 3: Development phase Gantt chart.

4.3 Changes to methodology

Thus far we have followed an Agile methodology, developing under two week long sprints. Every two weeks there would be a review of what is completed, and the new sprint focus that would be taken. This was fairly successful in constantly ensuring that research was always advancing, and that there was not too much time spent on covering one topic in depth. However, this

methodology failed in accounting for the lack of material that there was to evaluate on. There was no effective measure if the research undertaken was useful due to my unfamiliarity within the subject matter. More time should have been budgeted toward research, as well as time spent trying to make any crude solution using the tools and libraries found. This methodology resulted in very successful research to now be applied, yet it did cause delays in the project’s original timeline.

In the next phase of the project, the weaknesses of Agile will be less apparent, due to the heavier focus on development. Now that research has prominently been concluded, we will adapt the methodology such that each sprint only consists of a single week. Each sprint will have clear definable goals, producing measurable results that can be used to evaluate the success of the sprint and the project. This will encourage a greater incentive to develop faster in shorter windows - consistently producing code toward the project - but also enabling the project to adapt accordingly. The Gantt chart is a good time window to follow, however with our new strategy we aim to be more lenient on later weeks of the project, with external factors such as other university module deadlines, making us push work from one sprint ahead to the next.

5 Ethical discussion

Concerning generative NLP projects, there is always an important consideration to be made with what text the system will produce. Algorithms and AI have shown that depending on data they are trained on they can become biased or bigoted[51], and this is emphasised by LLMs like GPT-3.5[3]. LLMs sources of training are problematic for no guarantees of consistency for political leanings, which can result in responses to prompts to be biased. This bias can inadvertently cause censorship, or intentionally misdirect people. Bias isn’t the only risk, as the spread of misinformation from inaccuracy can cause disruptions to governments and democracy[18, 19], to spreading hate speech and inciting violence[4]. All of these are extreme cases, but when considering the aim of this project, if this app spreads misinformation, then it risks worsening the transparency between people and government. The risk of inaccuracy carries that of defamation and disrupting faith in the government; the app becoming a tool to mislead and misdirect people. Going forward, when using LLMs, we have to find some method of assuring accuracy, or at the very least ensuring to flag aspects of the summary that may not be accurate, to try and maintain that transparency. Otherwise this project may only contribute to what it is trying to reduce. In coming sprints of development, it will be important to conclude if it is even ethical to release an application like this, and if an assurance of accuracy is possible.

6 Conclusion

The progress of this project has been slower than originally intended, with such a large breadth of content regarding text simplification. While we are behind on the original plan, the research completed has established the foundations of the development phase of the project. We have powerful methods we can use, and techniques to measure our success; as well as a strong understanding of how we can improve on existing solutions. Utilizing LLMs and supervised learning techniques, we can overcome the ethical concerns and weaknesses of other approaches, to best simplify the UK’s laws and proposed bills.

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CS310 Project Specification

Mobile Application Enhancing Transparency of UK Politics

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1 Problem Breakdown

UK law, legislation and politics is difficult to understand. It is not a controversial statement to say that understanding the complexities and nuances of UK law and politics is no easy task. Reading any recently passed laws, you would struggle to easily decipher the meaning of the text. They are long, complex and time consuming to read. This is to the point that for any average member of the public, there is little incentive to see what MPs are actually achieving, rather just passively listening to proclamations and declarations from politicians - which poses risk of being full of half truths and misinformation. Hence, this has resulted in a decrease in trust between the UK populous and the government. Statistics released by the UK census for 2022 [4] revealed that only 35% of the population trusts the national government, and more interestingly that 58% of people are not confident people like them have a say in what the UK government does, otherwise known as poor political efficacy. These statistics alone indicate there is a gap between the public and the government that needs to be bridged. Both the OECD [7], and Grotlueschen's [2] data linked literacy levels to both trust in the government and political efficacy, with Grotlueschen's finding 60% of people with a literacy level of 1 assume poor political efficacy. This is a stark difference to only 30% of those with a higher literacy level assuming poor political efficacy. This makes a strong case that at least part of the reason that trust in the government is reduced is because a portion of the population cannot easily understand the more complex reading of UK law. Thus, it can be argued that we can start to mend the relationship by making the results of what MPs do, to be more widely accessible: by making legislation easier to understand.

There are two core problems facing the comprehension of laws and bills. The first issue is with respect to the language used. Reading an in progress bill, or an amended law, is difficult due to the number of references unknown to the reader. The text itself is too complex for the average reader. This cannot be easily proven as legislative texts cannot easily be assigned a reading level in any traditional sense, Looking at the PIAAC [6], they defined six levels of literacy proficiency levels, and reading their descriptions and examples, legislation would best fall into Level 4 or 5. From a 2012 study the PIAAC found the average level amongst UK adults was Level 2 [6]. Furthermore, 16.4% of adults in England have literacy levels below the National Literacy Trust's [8] Level 1, which is considered 'very poor literacy skills'. When also including that 8.9% of the population of England does not speak English as their first language [5], it paints a picture that a sizable portion of the population would not be able to grasp UK legislation. The second issue we face is the difficulty in remaining up to date with the legislative process. Bills are regularly introduced in the house of lords or commons, to either propose a new law, or to amend an existing one. The government does provide a website where you can access a database of every bill that has been proposed, and its progress [1]. However it is difficult to keep track of bills relevant to you, The solution to both of these problems is a mobile app that updates users regularly on bills and legislation that may affect them, and provides an interface to read a simplified or summarized version of the bill - ideally simplified to the average reading level of the country. This app aims to rectify the relationship between the public and the legislative process, by making it so ordinary people can understand what bills are being passed, what laws entail in a simpler form, and letting users stay up to date on what is relevant to them.

This previously would have been impossible to do, with no relevant existing technologies that would allow for such a project. However, improvements within natural language processing have made a substantial development, and shows new promise in both the realm of text simplification, and sentiment analysis. Text simplification has five key methods behind it, seq2Seq modelling [10], edit-based sentence simplification, lexical simplification, back translation, and structural sentence simplification [3]. Each of these are used in hybrid with each other, and have several models built for each of them. The best ones will need to be selected and trained for legislative texts. There is also certain evaluation metrics/training methods we will need to look at; a mathematical way to discern the complexity of the text, as certain methods prove unusable with legislation. We will also need to explore if the text produced may have any bias; which shall use sentiment analysis models to identify [9].

2 Objectives

The project has the overarching objective of providing a method for the simplification of UK legislation. This is the greatest priority and aim for the project, not only because it is the most technical and interesting aspect, but also because no application like it exists. Framing this as the most essential part of the project, sets up the rest of our objectives, and how we are to conduct the development process.

Breaking down the problem, developing the app divides into seven key areas to consider.

Key:

- (M) = Must do
- (S) = Should do
- (C) = Could do
- (E) = Extension

2.1 Module 1: Text simplification

A module that handles the simplification process. This module will have two major facets: bills and acts. When bills are being passed through the houses, they go through a series of phases, amending each step of the way until it is accepted and passed as an act. This leads to the fact that the final writing of bills and acts have different formats, and hence must each be dealt with accordingly. While the focus of the project is to ensure that any bill can be translated into something simpler, it is likewise important that acts and laws can be simplified as well. This is because some bills are amendments of existing laws, and because we need to demonstrate what the bill looks like once it has been passed.

Requirements:

1. The module can take in a text representation of a bill or an act, pass it into a text simplification model and return a simplified form of the bill or act. (M)
2. The text result from the model is accurate to the information inside of the text. (M)
3. The text result from the model can describe a series of impacts affecting certain demographics. (C)

2.2 Module 2: Political sentiment analysis

A module to evaluate the political sentiment of the translated text. This addresses the ethical risks discussed in the problem background. We will want to find a way to interpret what the political leaning of the text may be, and if it poses any potential risk of presenting bias. This module is less of a priority, as we need the previous module to be complete before this can be worked on.

Requirements:

1. A way for the system to identify if the given text simplification is appropriate to upload. (C)
2. A method for the system to analyze the political leanings of the given text using the left-right political spectrum. (C)
3. A method for the system to analyze the political leanings of the given text using the Eysenck political spectrum. (E)
4. A method for the system to evaluate if a given text is critical or favorable toward the focus subject matter. (C)

2.3 Module 3: Live updating system

A live tracking and update system. We will want to have a way to keep track of what bills are currently being passed through the houses, their current stage, their sponsor, and any other details relevant to that bill. This information can be accessed through the UK parliament website, however a retrieval, update and storage method will need to be designed.

Requirements:

1. A system in place to download and retrieve updated bill data, including short and long title, stage, sponsor, relevant dates, bill pdf. (S)
2. A method to format the downloaded data, into something efficient to store, and feed into the other modules. (S)
3. A system in place that adds to the simplified description of the given bill, expressing what has changed from the previous stage. (C)
4. A way of notifying relevant users to the update in data. (S)
5. An assurance the app is kept up to date and accurate to current information. (S)

2.4 Module 4: Demographic identification

A method to identify what legislative texts impact what demographics. We will need to identify a series of demographics that is relevant to both the users and to the texts. This likely will be an extension of the first module, utilizing the model to find these demographics in its answer.

Requirements:

1. A method to identify what demographics are impacted by an introduced bill or an act. This demographic will be pulled from a collection of predefined demographics. (C)
2. Research into all relevant/important demographics that need to be addressed and covered by the model. (S)

2.5 Module 5: Database

An SQL database that can let us store user data, and information about the bills and legislation. The user data would be entered through self identification, designating details such as age, gender, occupation, tax bracket, etcetera. This would then need to be somehow securely stored, so as to not violate GDPR. The database storing information on the bills and legislation meanwhile will want to contain the simplified version of the legislative text, so that it is consistent in meaning and quality when distributed amongst the users. Additionally we will want to store any relevant information to the text that we would want it to display. The reason the database will be SQL is because we will want to make several complex queries surrounding the users impacted by certain demographics, and the bills that individual users may be tracking.

Requirements:

1. A well structured normalized database which contains all relevant details to be used by all other modules. (S)
2. Storing self-identified user data. This includes personal details such as: age, occupation, gender etcetera. (S)
3. Ensuring security for the storage of given user data. (M)
4. Storing important relevant information about a given bill or act. Storing the simplified text rather than the original document. (S)
5. Designed such that it can be expanded to introduce new relevant details. (C)
6. Assurances that user data will be securely stored, and will not violate GDPR. (M)

2.6 Module 6: UI

An engaging and accessible UI for users on the phone app. This project is with the aim of making legislation more accessible to users, and part of that is making it not boring. Reading legislative texts is incredibly boring, and this product is aimed at alleviating that, hence if the presentation of the UI and app is more drab than the text, it will fail the project's primary objective. The interface must be both interactive, simple and engaging, taking many attributes from modern social media apps.

Requirements:

1. Mobile app user interface designed, such that it is functional, and all elements of the interface act as expected. (S)
2. A method to ensure to write the text data from the database about the legislation, without any display errors. (S)
3. A way to access the original bill from inside of the app. (E)
4. Easy to use, engaging and interactive for the users. (S)

2.7 Module 7: Ethical discussion

Reaching a conclusion on the place of AI technologies in law and politics. Throughout this project, especially with regards to the legislative text simplification and political sentiment analysis, there will be much research put into how to design the algorithms and models so that there will be no inaccuracy or bias. However, there is no guarantee of success for either front. This project is using its given product as a medium to investigate the future of this technology, and concluding if it's ever moral or ethical to use technology in this way.

Requirements:

1. Research the use of technology, AI, and automation in the realm of law and politics. (S)
2. Evaluate if it is ethical to use these technologies in law and politics. (S)
3. Evaluate if it can ever be safe to use these technologies in law and politics. (S)
4. Evaluate if AI technologies can ever truly be unbiased in politics. (S)
5. Use evidence of existing case studies, and use findings from the results of this project's model and technologies. (S)

2.8 Other Extensions:

1. An ability for the user to leave comments on a given bill or act simplification. (E)
2. An ability for users to vote/request new simplifications on given bills or acts. (E)
3. Additional information and resources on the legislation page produced by third party sources, such as ground news. (E)
4. An educational opening which informs the user in simpler terms how the legislative process actually works. (E)
5. Make the app more accessible to a larger number of people, with accessibility settings such as text size, audio prompts and readings, and in-built definitions for words which may not be as easily understood by all users. (E)

3 Methodology

The methodology used will be Agile. Agile allows us to bring the most important part of the project, and that is the flexibility. Due to how much research and experimentation that this project involves, it is incredibly likely that a goal or an approach will have to change to accommodate this, especially since certain modules are highly dependent on other modules working. Over the course of the next 7 months, I will be undertaking 2 week long sprints, where each sprint will focus on research, implementation, development and then testing. At first the iterations will focus on designing and improving module 1, the text simplifications but as the project develops, we will branch out the focus into the other modules accordingly. This process will involve a lot of reevaluation stages, where the we will refocus time and resources on to the respective modules that need it the most. Modules are functionally independent from each other, and theoretically each can be done on their own, which leads to what needs to be prioritised. Ordering the modules in priority of highest to lowest is: 1, 3, 5, 6, 4, 2. Modules will likely be done in parallel as the project progresses, needing to adapt to requirement amendments during reevaluation. Once the first module is functional enough for the project to progress, parallel development will begin.

4 Timetable

For the timetable, you can see in the Gantt chart below, the plan I intend to follow for term 1. This term will focus primarily on module 1, 3 and 5. This set up is so that it focuses on the priority modules for the software, and then allows term 2 to focus on the development of the UI, the demographic identification, and if there is time, the political sentiment evaluation. This has been done to avoid scope creep, as there are many unknowns with this project, and by following this timetable, we will maximise the effectiveness of what we complete.

The first two sprints (starting from week 3) focus on module 1, and aims to have it working to some extent by the end of the second sprint. This opens the door for focus on module 3 and 5. Module 6 is begun at the end of term, however with term 1 coursework for other modules being due at that time, it may not receive much work until after the term has ended, hopefully leading into after Christmas.

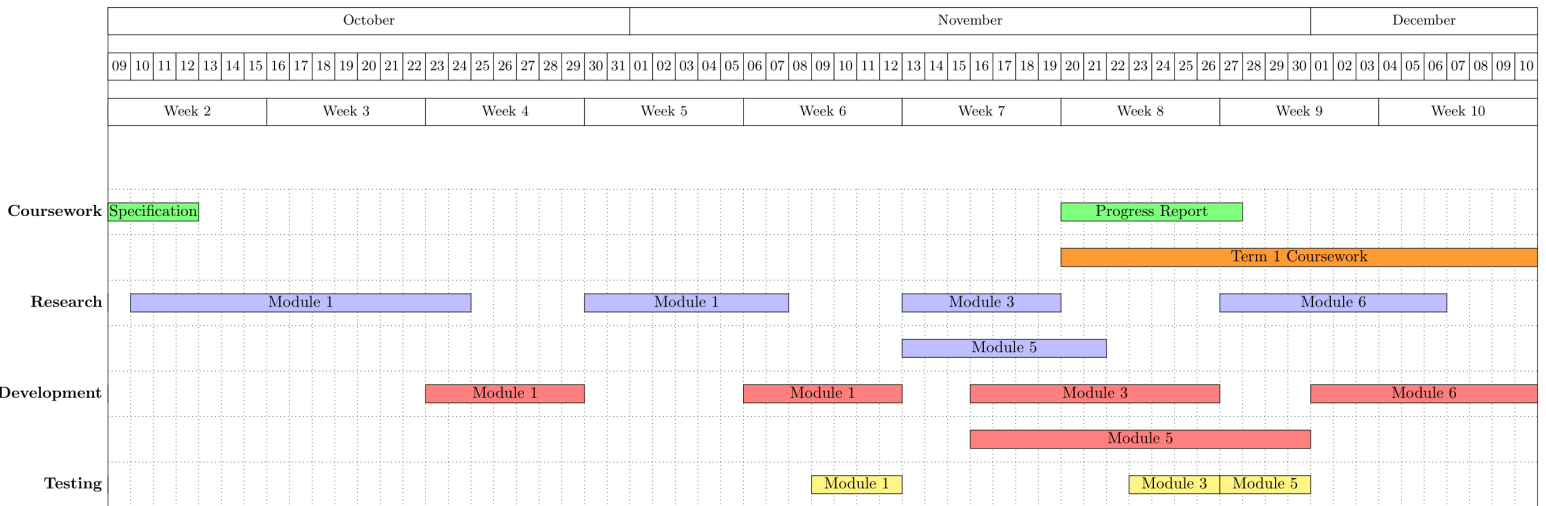


Figure 1: Term 1 Gantt Chart

5 Resources

Initial known resources for the project. May change or expand during project development.

- GitHub: This will be used for version control of the code as it is developed.
- Text simplification models: There are many existing models that may suit my purposes. These include Hugging Face, PyTorch (Seq2Seq), Tensor2Tensor, EditNTS, LSBert, Glavas, and LaserTagger.
- Text datasets: These will be useful in simplification, however available ones I have found may not best account for legislative dialogue and will need expansion. These datasets may still find some use though: X-Sum, D-Wikipedia and Newsela.
- Political sentiment models: There are less models that look at this, but two which show promise are PyABSA, and Textblob. This will definitely be expanded upon as the project develops.
- UK parliament website: This will be used to access every legislative document used.
- Mobile app development tool: An app development tool which will be used for development; will need to be able to make calls to other modules, and the database.
- SQL database server: Some method of implementing a server to upload to and query from across devices, likely using AWS or Azure.

6 Risks

Here are the given risks for the project, and how their impact will be reduced or mitigated:

- Other courseworks or deadlines getting in the way:
Ensuring there is enough redundancy in my time management.
- Difficulty learning mobile app development:
Ensuring there is enough redundancy in my time management. Worst case scenario switching to web development if upon evaluation appears too difficult.
- Government website goes down and cannot access their files:
Download an appropriate subset of files at random that can be used for use and testing.
- Lack of storage space for all legislation document details:
Research into the use of external storage software, such as cloud, or through the department.
- Development PC suffers damage such that the files on it are inaccessible:
GitHub will be regularly updated with the project and its resources.

- Bias within the system. There is a real risk that the model or training data used, could accidentally skew the simplifications to certain political affiliations. This risks radicalising people, which can lead to increased extremist actions, or even inspire criminal activity against the government:
Attempt to implement a political sentiment analysis tool, to prevent this from having extreme bias. It will not be released to the public until some assurance of no extreme bias. This will need an independent review during the testing process by myself to ensure the text produced is suitable.

7 Legal, Professional and Ethical Considerations

There are a couple legal considerations, and importantly ethical issues that need to be discussed.

Legal issues:

- GDPR concerns: I need to make sure that any personal data stored by a user is securely stored inside of the database. Alternatively, all data provided by the user is fictitious. This will depend on the development of the project and will be reevaluated.
- Defamation risks: This is a risk of information spread by the app is incorrect. This inherently doesn't pose any legal risks, however during testing it will be important to inform the testers that information may not be accurate. Hopefully as development progresses, methods to avoid this risk will be researched and found.

Ethical issues:

The risks of accuracy and bias, within the system and outside of it, leads to an uncomfortable ethical question with regards to our recent advances in technology. Should we use automated AI systems, within the realm of law and politics? The issue arises from bias, as there is evidence that AI can be trained on bias data, which can result in the model itself being biased with dangerous impacts on groups and individuals. This product may influence the populous negatively with a biased take on the legislation. Furthermore, if this product was released publicly, maintained and run by a person, business or organization, then this tool could be used to intentionally skew public opinion on certain bills and laws. This would inadvertently turn the product into a tool for propaganda. This feeds into the question of if it is possible to have an unbiased system, especially with something so subjective as politics. This project will aim to explore what it means to use AI, and to a large scope, technology, inside of politics; questioning if it is possible to create something that is consistently unbiased.

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